Natural Language Processing Group

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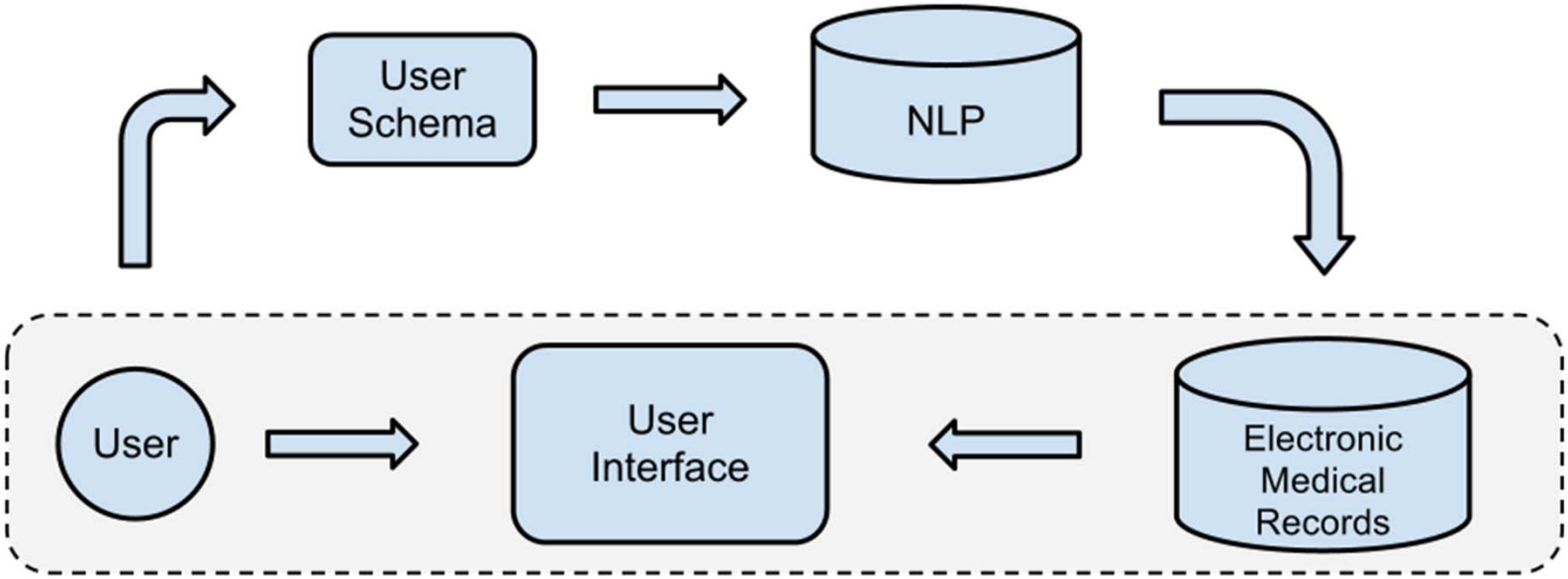
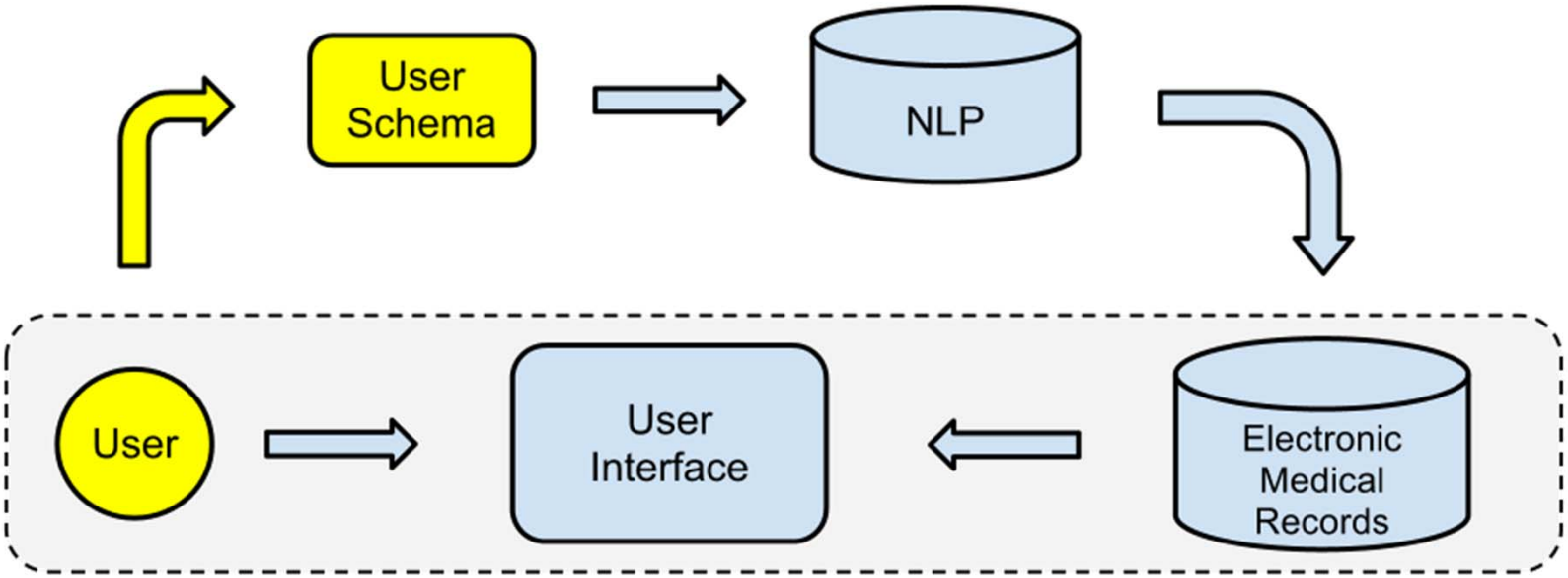
Natural Language Processing



Schema Builder



User Studies for Schema Builder

- Supervised by Wendy Chapman, Ph.D.
- Matthew K. Hong
  - » Human-Computer Interaction, UCSD
- Project Duration
  - » 6.18 – 9.18

Agenda

- Motivation for Schema Builder
- User Studies
- Findings from User Studies
- Mock-ups of Schema Builder

Why is user schema important?

- Problems with clinical data
  - » Too much information
  - » Unstructured
  - » Not all information is relevant

- Schema Builder
  - » Allows user to build and customize NLP schema for extracting and structuring relevant information

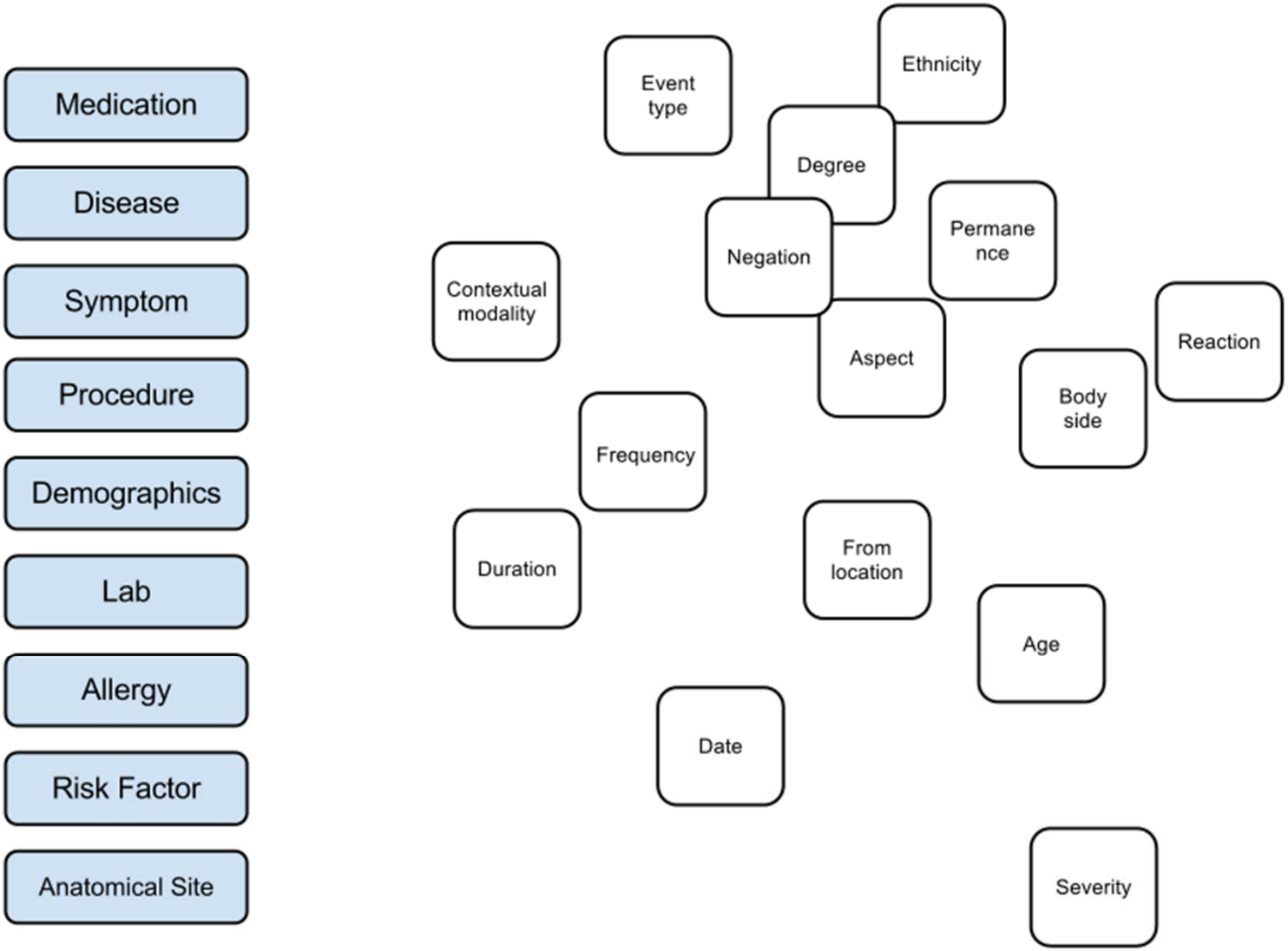
iDASH

Understanding the user is important

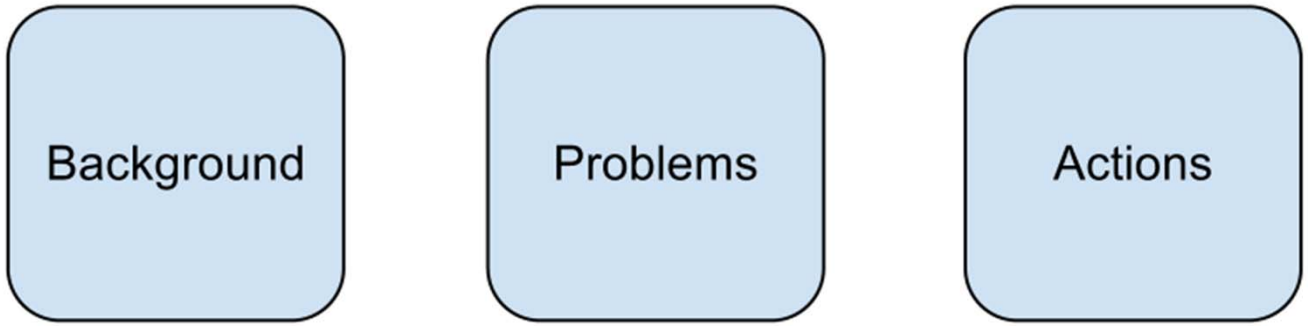
- Understanding the user
  - » Who will use SB?
  - » What do they know?
  - » What do they want to see?

- User Studies
  - » 30 minute task-oriented interviews (3 sessions)
  - » 3 use cases
    - Adverse Drug Events
    - Colonoscopy Exam
    - Bleeding events (future)

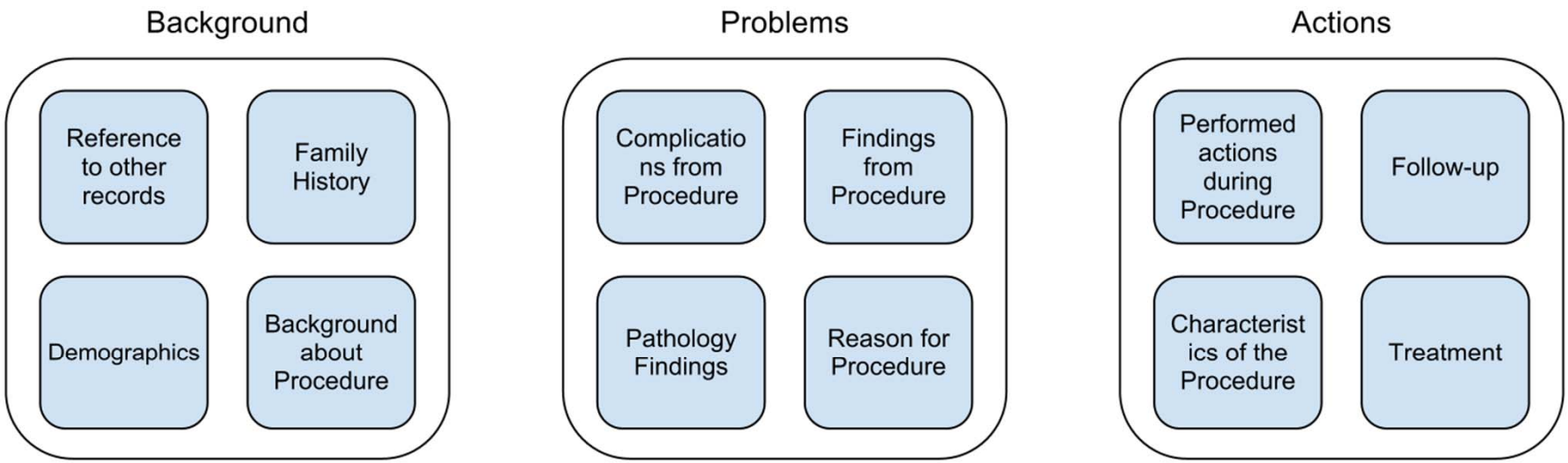
Task: Mapping concepts and attributes



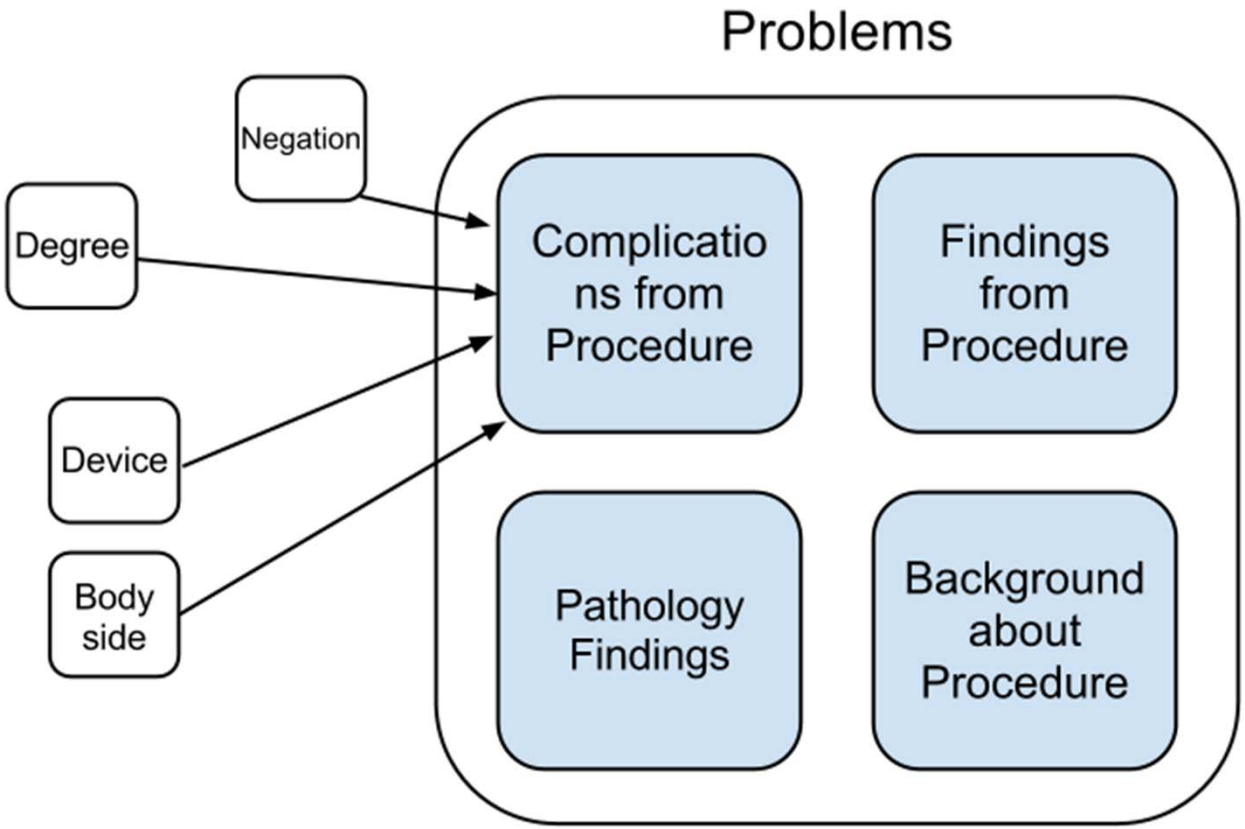
Findings: Three Categories of Concepts



Subcategories



Assigning Attributes



User Display of NLP Output

- Colonoscopy use case
  - » Form
- Adverse drug events
  - » Flow diagram (interactive)
  - » Timeline

Form

Input Form

Data Collection Form

Report # colonoscopy | Coder

Background

Report references nursing reports: Yes / Not Recorded

Family history of colorectal cancer: Yes / No / Not Recorded

Indication Type

Indication Type_2

Indication Type_3

No indication provided

Previous colonoscopy (yrs) | Not Recorded | ASA | Not Recorded

Informed Consent: Yes / No / Not Recorded

Basic Procedure

Withdraw_time (min) | Not Recorded

Procedure aborted: Yes / No/Not Recorded

Prep adequate: Yes / No / Not Recorded

Reached the cecum: Yes / No / Not Recorded

Saw appendiceal orfice: Yes / No / Not Recorded

Saw ileo-cecal valve: Yes / No / Not Recorded

Any complication?: Yes / No / Not Recorded

Complications: Bleed arterial/spurting; Consult/admission/tranfusion; Perforation; Bleed epi/ clips/cautery; Resp dep/ arrythmia

Findings - Polyp

No polyp; Polyp size not indicated

Polyp_size_largest (mm) *Convert to MM*

Any polyp removal?: Yes / No

Piecemeal removal of polyp?: Yes / No

Findings - Biopsy

Was any biopsy done?: Yes / No

If biopsy obtained, was location indicated?: Yes / No

Follow-up

Follow-up time (yrs) | Not Recorded

If follow-up time indicated, is caveat given?: Yes / No

Notes that you want to add

Save Record

Pathology Report (see pt id to match)

Pathology Report # | No pathology report

Any adenoma: Yes / No/Not recorded

Largest adenoma size (mm) | None

Convert all sizes to MM

Polyp size path largest (mm) | None

Bad pathology?: Yes / No

Record: 1 of 1 | No Filter | Search

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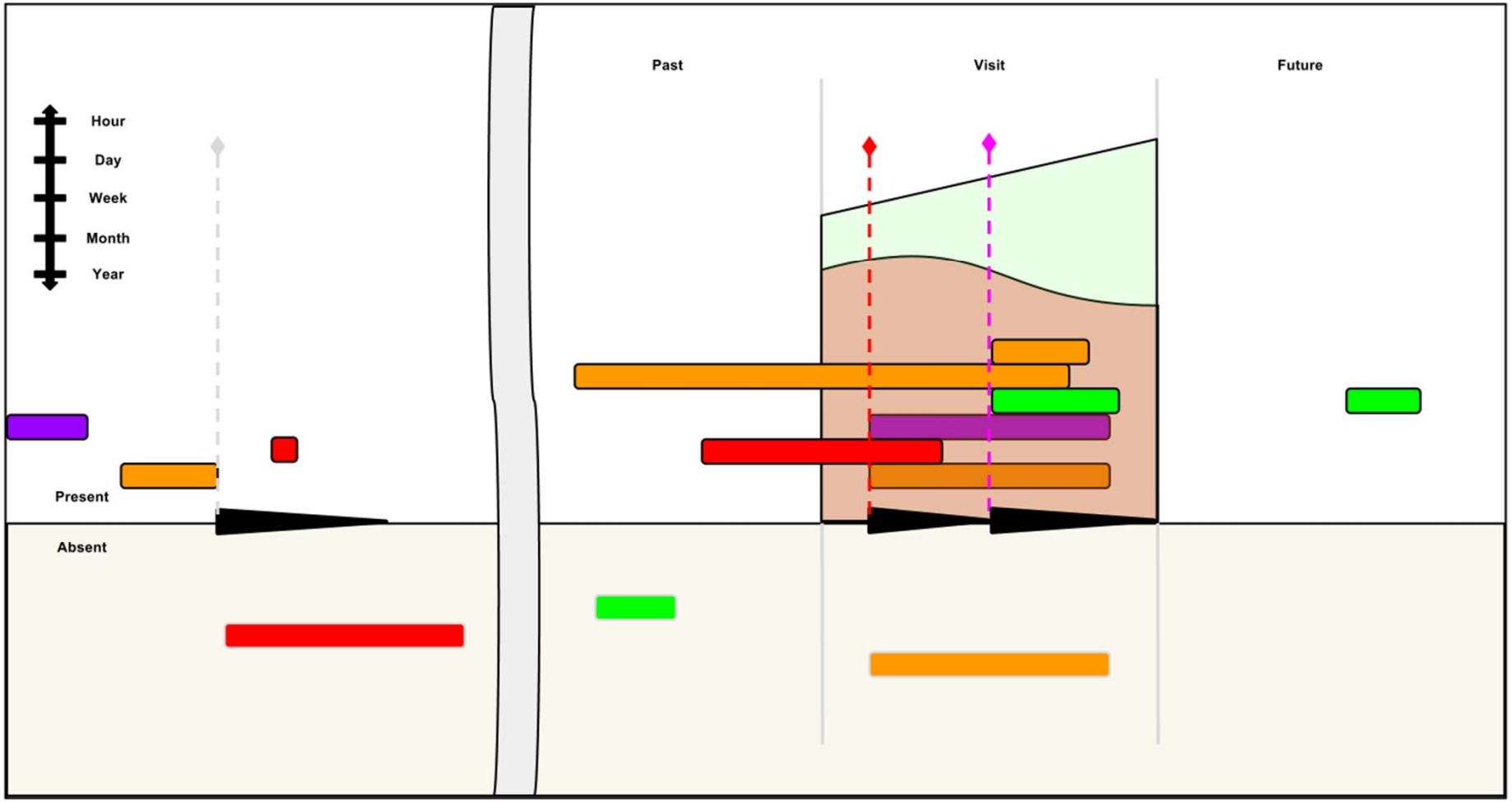
Timeline Visualization



Summary and future directions

- Lessons learned
  - » What categories of information is important?
  - » What attributes are relevant?
  - » How can we display the NLP-extracted information to assist in workflow of research study.
- Future plans
  - » Iterative design
  - » Linking NLP system to the user display
    - NLP automatically fills in the information
    - User can drill down to the source

Acknowledgments

- Users
  - » Ateev Mehrotra, Sandra Kane-Gill
- Collaborators
  - » Melissa Tharp, Harry Hochheiser, Rebecca Hwa, Jan Wiebe
- Calit2
- iDASH and DBMI
- NIH (NLM)

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